Slow Dynamics and High Variability in Clustered Networks

Reproduction of Litwin-Kumar and Doiron (2012)

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Abstract

This report reproduces the key findings of Litwin-Kumar and Doiron (2012) on slow dynamics in clustered cortical networks. We implement a network of leaky-integrate-and-fire neurons with increased connection probabilities based on neurons cluster assignment. We examine the effects of clustering on the network's firing patterns through measures like Fano factor, correlation coefficients, and firing rate distributions.

1. Introduction

Cortical networks are often modeled as unstructured, randomly connected networks with balanced excitation and inhibition. This balance can place the network in an irregular, asynchronous regime in which neurons exhibit high trial-to-trial variability in spike timing. In this regime, excitation and inhibition are large but cancel in the mean. Thus, spikes are often fluctuation-driven, resulting in high spike-time variability.

However, cortical networks also exhibit slower, population-level changes in firing rate over longer timescales (hundreds of milliseconds to seconds). Baseline, unstructured, balanced excitation-inhibition networks do not typically capture these slower fluctuations in firing rate.

In this study, we find that introducing clustered excitatory connectivity creates slow dynamics. Clusters transition between states of high and low firing, such that neurons exhibit both high spike-time variability and slow firing-rate fluctuations. Introducing stimuli can bias the network toward particular activity states, reducing variability.

2. Results

We present our reproduction of the key findings from the original study.

2.1. Homogeneous Network Baseline

Networks without clustering exhibit stationary activity with low variability.

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2.2. Clustered Network Dynamics

Introducing clustering produces slow fluctuations through transient cluster activations.

2.3. Quantitative Measures

We quantify temporal structure using Fano factor and autocorrelation analysis.

3. Discussion

Our reproduction confirms that clustered connectivity generates slow dynamics through metastable attractor states.

3.1. Interpretation

The clustering mechanism provides a parsimonious explanation for slow cortical fluctuations.

3.2. Limitations

The model uses simplified neurons and imposed rather than emergent clustering.

3.3. Future Directions

Promising extensions include plasticity mechanisms and functional implications.

4. Extensions

Beyond reproduction, we explore extensions to the clustered network model.

4.1. Extension 1

Description of the first extension and its motivation.

4.2. Extension 2

Description of the second extension and its motivation.

5. Model

We implement a recurrent spiking network with clustered excitatory connectivity using Brian2.

5.1. Network Architecture

The network consists of excitatory and inhibitory leaky integrate-and-fire neurons with structured connectivity.

5.2. Neuron Model

Membrane potential dynamics follow a leaky integrate-and-fire model with exponential synaptic filtering.

5.3. Connectivity

Connection probability between excitatory neurons depends on cluster membership, with enhanced within-cluster connectivity.

5.4. Parameters

Model parameters are chosen to match those of the original study.

References